## 1. Name of Project

Design and Implementation of an Automatic Overcurrent Protection System using Arduino.

## 2. Abstract

Overcurrent conditions in electrical systems can damage equipment, reduce system reliability and create safety hazards. This project presents an AC overcurrent protection system using an ACS712 current sensor, Arduino microcontroller and MOC3021 optocoupler. The ACS712 sensor continuously measures the current drawn by a 220 V, 50 Hz AC load. The Arduino processes the sensor output and calculates the RMS current value. When the measured RMS current exceeds the preset threshold of 0.5 A, the system enters a protection mode and disables the load by changing the output state connected to the MOC3021 optocoupler. The protection state remains latched even if the current returns to a safe level. A push button is used to manually reset the system and restore normal operation. The use of the MOC3021 provides electrical isolation between the low-voltage control circuit and the high-voltage AC mains, improving operational safety. This project demonstrates practical applications of current sensing, signal processing, fault detection and electrical protection.

## 3. Circuit Diagram

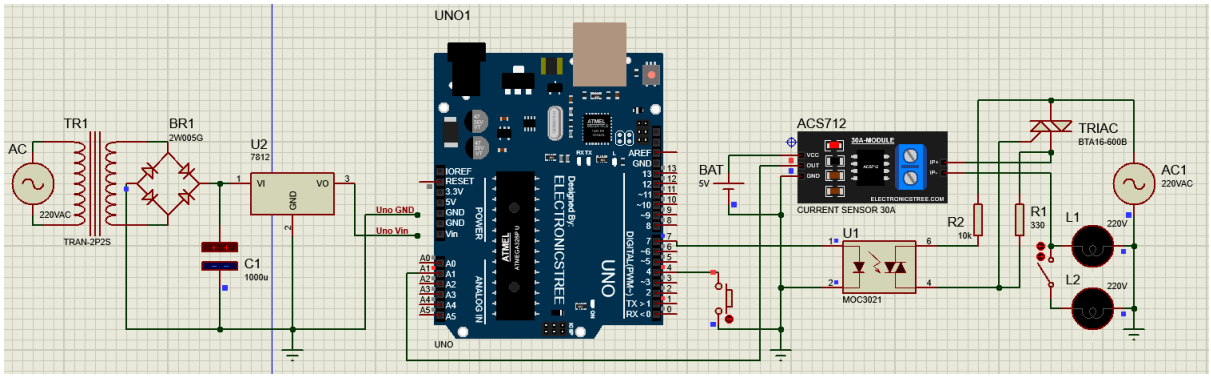



**Fig. 1:** Circuit Diagram of an Automatic OCP System

## 4. Component Description

### *4.1 Arduino Microcontroller*

The Arduino serves as the central controller of the system. It receives analog data from the ACS712 current sensor, calculates the RMS current, compares it with the predefined threshold and controls the protection output through Pin 7. It also monitors the reset push button connected to Pin 4.

### *4.2 ACS712 Current Sensor (30 A)*

The ACS712 is a Hall-effect-based current sensor capable of measuring both AC and DC currents while providing electrical isolation between the sensing path and output signal. The 30 A version of the sensor is used in this project, having a sensitivity of approximately 66 mV/A. The sensor produces an analog voltage proportional to the current flowing through the load, which is read by the Arduino through analog pin A1.

### *4.3 BTA16-600B TRIAC*

The BTA16-600B is a TRIAC used to switch 220 V AC loads. It is triggered through the MOC3021 optocoupler and acts as the main power switch. When an overcurrent condition is detected, the trigger signal is removed, turning OFF the TRIAC and disconnecting the load.

### *4.4 MOC3021 Optocoupler*

The MOC3021 is an optically isolated TRIAC driver used to provide electrical isolation between the Arduino and the high-voltage AC circuit. Pin 7 of the Arduino is connected to the input LED of the MOC3021. When the Arduino output activates the optocoupler, the internal photo-triac triggers the BTA16-600B TRIAC, which controls the AC load.

During normal operation, the Arduino maintains the HIGH output state on Pin-7 to allow the AC load to operate. When an overcurrent condition is detected, the Arduino changes the state of Pin 7 to LOW, causing the MOC3021 to stop triggering the power TRIAC. As a result, power to the load is interrupted, protecting the circuit from excessive current.

### *4.5 Step-Down Transformer (220 V/12 V)*

The 220 V/12 V step-down transformer reduces the 220 V AC mains voltage to 12 V AC and provides electrical isolation. The 12 V AC output is supplied to the bridge rectifier for DC power generation.

### *4.6 Diodes (1N5408)*

The 1N5408 is a high-current rectifier diode rated for up to 3 A and 1000 V. In this project, four diodes form a bridge rectifier that converts 12 V AC into pulsating DC. The rectified output is then filtered and regulated to supply the control circuit.

### *4.7 Voltage Regulator (LM7812)*

The LM7812 is a fixed positive voltage regulator that provides a stable 12 V DC output. It regulates the filtered DC voltage and supplies a constant voltage to the control and switching circuit for reliable operation.

### *4.8 Capacitor*

A 1000 µF electrolytic capacitor is connected after the bridge rectifier to smooth the pulsating DC voltage. It reduces ripple and provides a stable input to the LM7812 voltage regulator. Another 1 µF capacitor is connected between the sensor output and ground to reduce high-frequency noise and stabilize the sensor signal.

### *4.9 AC Load*

The AC load represents the electrical appliance or group of appliances connected to the 220 V, 50 Hz supply. Two incandescent light bulbs of 60 and 100 Watts were used.

### *4.10 Push Button*

A push button is connected to Pin 4 of the Arduino and configured using the internal pull-up resistor. The button functions as a manual reset switch. After an overcurrent trip occurs, pressing the button allows the system to resume normal operation.

## 5. Circuit Operation

To validate the performance of the protection system, two incandescent light bulbs rated at **60 W** and **100 W** were used as the AC load, connected in parallel.

The system continuously monitors the current flowing through the AC load using the ACS712 current sensor. The sensor output is connected to analog pin A1 of the Arduino. Since the load operates on 220 V AC, the Arduino samples the sensor output multiple times and calculates the RMS current value to determine the effective current flowing through the load.

Initially, only the **60 W bulb** was turned ON. Under this condition, the ACS712 sensor measured an RMS current of approximately **0.3 A**, and the system operated normally without triggering any protection. Pin 7 remained HIGH, allowing the MOC3021 to activate the TRIAC and keep the load powered.

Next, the **100 W bulb** was switched ON in parallel with the 60 W bulb. This increased the total load current significantly, and the measured RMS current rose to approximately **0.78 A**. Since this exceeded the predefined threshold of **0.5 A**, the Arduino immediately detected an overcurrent condition and activated the protection mechanism.

Consequently, Pin 7 was driven LOW, disabling the MOC3021 optocoupler and stopping the triggering signal to the BTA16-600B TRIAC. This resulted in disconnection of the AC supply to the load, turning OFF both bulbs. The system remained in this tripped state until the reset push button on Pin 4 was pressed, after which normal operation resumed.

This demonstrates a reliable, fast, and latched overcurrent protection system suitable for AC load safety applications.

## 6. Cost Calculation

| Sl. | Items | Qty. | Price |
|---|---|---|---|
| 1 | 1N5408 Diode | 4pcs. | 10 |
| 2 | MOC3021 | 1pcs. | 40 |
| 3 | 7812 Voltage Regulator | 1pcs. | 15 |
| 4 | BTA16-600B | 1pcs. | 20 |
| 5 | ACS712 | 1pcs. | 150 |
| 6 | Bulbs – 60W + 100W | 2pcs. | 75 |
| 7 | Holder | 2pcs. | 70 |
| 8 | 220V/12V Step Down Transformer | 1pcs. | 200 |
| 9 | Capacitor - 1000uF & 1uF | 2pcs. | 30 |
| 10 | Plugs | 2pcs. | 40 |
| 11 | Arduino | 1pcs. | 800 |
| 12 | Terminals | 2pcs. | 20 |
| 13 | PCB Board | 1pcs. | 70 |
| 14 | Bed Switch | 1pcs. | 25 |
| 15 | PVC Board – 3mm | - | 40 |
| 16 | AC Cables | - | 30 |
| **Total** | | | **1635** |

## 7. Project Implementation Video Link

Demonstration video: assets/video/demo.mp4 (included in this repository).

## 8. Project Picture

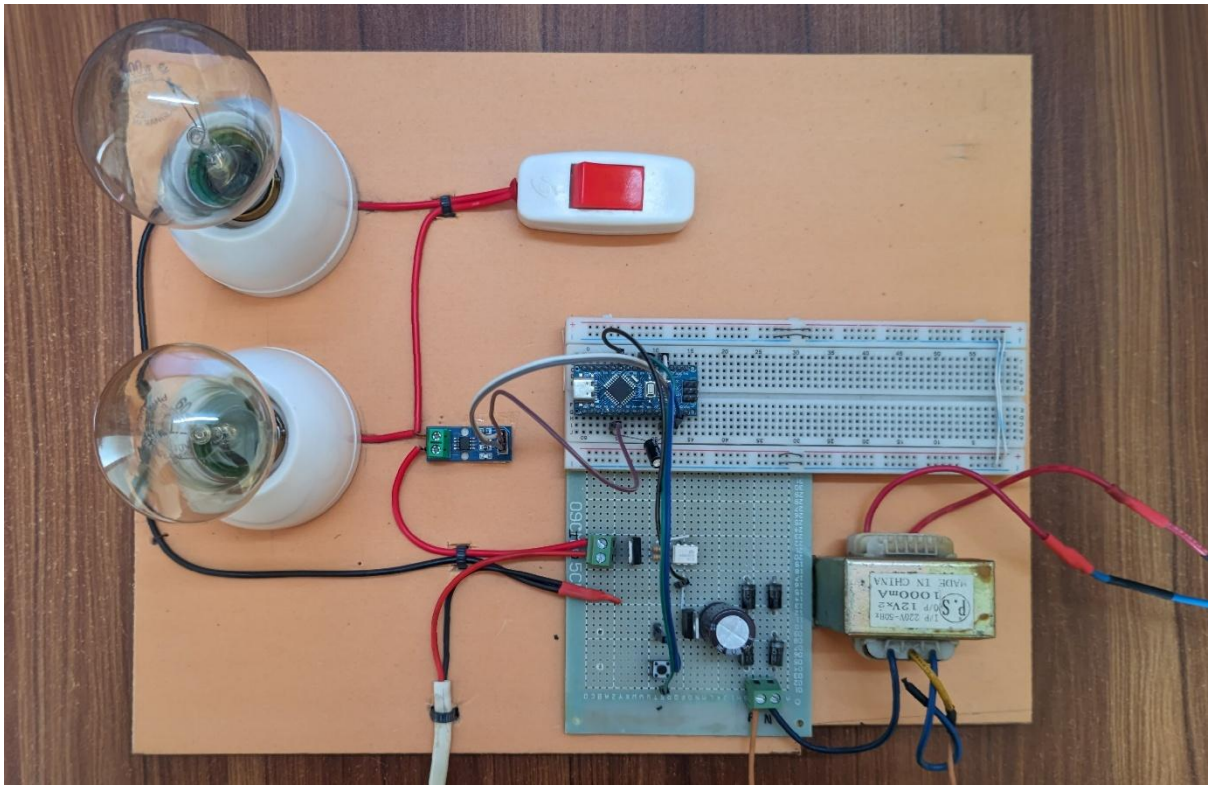


**Fig. 2:** Hardware Implementation of an Automatic OCP System